

Journal Entry: Current Project Status

calc_mbcwc

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Status

The project now has a working first benchmark target for the Stanford matrix Φ^4 Schwinger-Keldysh calculation. The core result is implemented in `scripts/stanford_phi4_kernel.py` and tested in `tests/test_stanford_phi4_kernel.py`.

The benchmark is model driven. It starts from the Stanford matrix Φ^4 model, applies the standard Schwinger-Keldysh branch lift

$$\phi_1 = \phi_r + \frac{1}{2}\phi_a, \quad \phi_2 = \phi_r - \frac{1}{2}\phi_a,$$

derives the normalized branch-difference coefficients of

$$\phi_1^4 - \phi_2^4,$$

records the on-shell ladder-kernel equations from the Stanford paper, and evaluates the leading fixed-mass and thermal-mass-improved massless behavior.

Recovered Stanford Answers

The current pure-Python benchmark reproduces

$$m_{\text{th}}^2 = \frac{2\lambda}{3\beta^2}, \tag{1}$$

$$\lambda_L \simeq 0.025 \frac{\lambda^2}{\beta^2 m}, \quad m\beta \ll 1, \tag{2}$$

$$\lambda_L \simeq 0.031 \frac{\lambda^{3/2}}{\beta}, \quad m = 0, \tag{3}$$

where the massless expression uses the thermal-mass improvement.

The Appendix kernel diagonalization also reproduces the small-mass coefficient numerically. For the tested $m\beta = 0.1$ grid,

$$\frac{\beta^2 m}{\lambda^2} \lambda_L = 0.0256986096,$$

consistent with the quoted Stanford value near 0.025.

Math Audit

The math audit found and corrected important numerical details. The diagonal $E_- = 0$ case in the Appendix kernel now uses the analytic limit rather than an artificial epsilon. The inverse hyperbolic-sine factors are evaluated stably at large argument. The default momentum mapping scale P_0 is now mass-regime aware, using the Stanford recommendation $P_0 \approx 3m$ in the small-mass regime and $P_0 \approx m$ otherwise. The numerical matrix is checked for symmetry before using a symmetric eigensolver.

The symbolic audit clarified that the quantities previously called SK vertex weights are normalized branch-difference coefficients only. They do not include the interaction sign, coupling, or real-time factor required for full Feynman vertex factors. The code now exposes that distinction in the pipeline payload.

Verification

The current verification evidence is:

- `python3 -m unittest tests/test_stanford_phi4_kernel.py`: 20 tests pass.
- A scoped non-Wolfram regression suite: 51 tests pass.
- Full discovery still has known `WolframKernel` metadata-export segfaults in the Wolfram-dependent tests. Those failures are separate from the pure-Python Stanford benchmark and remain a runtime stability issue to solve later.

Next Target

The next research target is to connect the benchmark pipeline to a general model-to-Schwinger-Keldysh-kernel pipeline. The current Stanford implementation is an acceptance reference: future generalization should keep reproducing these answers automatically while replacing Stanford-specific assumptions with data derived from an input model.